Amazon Inventory Reconciliation Using AI: Counting Items in Bin Images

UE24CS352A Machine Learning, Mini-Project | Tejas Shetty (PES1UG24AM305) · Tanishk Gowda (PES1UG24AM301)
Code: github.com/<user>/amazon-bin-reconciliation (private, shared with faculty)

1. Problem statement

In Amazon fulfilment centres, robots carry pods of bins and items are stowed into bins at random. Items are occasionally misplaced, so the inventory system's recorded count no longer matches what is physically in the bin. Given only an RGB photo of a bin, we predict the number of items (0 to 5) and use that prediction to decide whether the bin agrees with its recorded quantity or should be sent for a manual audit. We reproduce and extend the Stanford CS229 study by Rodriguez Bertorello et al., which compared SVMs with ResNet CNNs.

2. Dataset

The public **Amazon Bin Image Dataset** (AWS Open Data, about 535K bin photos) pairs each JPEG with a JSON whose EXPECTED_QUANTITY field is our label. Following the paper, we keep bins with at most 5 items, giving a 6-class problem with a 16.7% random baseline. Our downloader samples random bins from the public S3 bucket (default 20,000 usable images, resized to 256x256), and we use a stratified 70/20/10 train/validation/test split. The classes are imbalanced (empty bins are rare), photos vary in lighting, and many items are hidden behind the elastic tape that holds them in place, so some labels cannot be verified even by a human.

3. Approach

Classical baselines. Images are down-sampled to 64x64 and described by (a) raw pixels reduced with PCA (150 components, randomized solver), (b) HOG on grayscale and on a Sobel edge map (to suppress tape stripes), (c) Laplacian-of-Gaussian blob statistics (count, radii, spatial spread of bright and dark blobs) and (d) colour histograms. We train Logistic Regression, a Decision Tree (Gini), a linear SVM and an RBF-kernel SVM, with standardisation and class-balanced weights, tuning C, gamma and depth by stratified 5-fold GridSearchCV on the training split.

CNNs with transfer learning. ImageNet-pretrained ResNet18, ResNet34 and EfficientNet-B0 get a new 6-way head (dropout + linear) trained with cross-entropy loss. Training has two phases: (1) a warm-up epoch with the backbone frozen so the randomly initialised head does not damage pretrained features, then (2) full fine-tuning with discriminative learning rates (lr/9 for early layers, lr/3 for middle layers, lr for the head). We compare SGD with step decay (the paper's baseline), SGD with warm restarts (SGDR), and AdamW with a one-cycle schedule, with the initial learning rate set by a learning-rate range test. Augmentation is kept mild (horizontal flip, small affine shift, colour jitter), because aggressive crops could cut an item out and silently change the label. We use optional inverse-frequency class weights, mixed precision, early stopping on validation accuracy, and flip test-time augmentation.

From counting to reconciliation (our extension). The business question is whether a bin matches its record, not the exact count. We flag a bin when the model's probability for the *recorded* count, P(count = recorded), falls below a threshold. To evaluate this, we simulate an inventory system in which 50% of records are wrong (mostly off by one, as when a single item is misplaced) and report ROC-AUC, precision and recall of the flag. Since accuracy alone hides how wrong a count is, we also report RMSE, MAE, off-by-one accuracy and macro-F1.

4. Implementation

The project is written in Python with scikit-learn and scikit-image for the classical models and PyTorch and torchvision for the CNNs. Its modules are: a resumable parallel S3 downloader, a stratified splitter with EDA, a cached feature extractor, GridSearchCV training, a two-phase CNN trainer with an LR finder and four schedulers, an evaluator that produces a confusion matrix and a gallery of the most confident mistakes, and a results summariser. A **Gradio demo** takes a bin photo and the recorded quantity and returns the count distribution, a MATCH / FLAG FOR AUDIT decision and a **Grad-CAM** heat-map of the regions behind the prediction. Seeds are fixed, the whole pipeline runs with one script (PowerShell, bash or a Colab notebook), and pytest smoke tests on a synthetic dataset check every stage end to end.

5. Results (held-out test split)

Model	Family	Test acc.	Off-by-one	RMSE	Macro-F1	Recon. AUC
[run experiments, then re-run this script]						
Paper: ResNet34+Adam, 324K imgs (reported)	ref.	56.2%	-	0.900	-	-
Paper: SVM RBF + PCA (reported)	ref.	32.0%	-	-	-	-
Random guess (6 classes)	ref.	16.7%	-	-	-	-

Paper rows are reported numbers on a different sample and split, shown for context only. Recon. AUC = mismatch-detection ROC-AUC from P(count = recorded).

6. Conclusions

[Auto-filled after experiments: best CNN accuracy vs. best classical model and vs. random; per-class trend; reconciliation AUC/precision/recall; comparison with the paper.]

Qualitative findings to confirm against your own error gallery: accuracy is highest for empty bins and falls as the count grows; most errors are off by one; the hardest images have items hidden behind the elastic tape or stacked on each other, which even a human cannot count reliably.

Limitations and future work. Labels are noisy when items are fully occluded, and the 0-5 subset excludes fuller bins. Promising next steps are cleaning or down-weighting unverifiable images, training on more of the 535K images, ordinal or regression losses that penalise large count errors more, and ensembling architectures.

7. Individual contributions

Tejas Shetty: data pipeline and EDA, feature engineering (HOG, LoG blobs, PCA), classical baselines and hyper-parameter search, evaluation metrics including the reconciliation simulation, write-up. **Tanishk Gowda:** CNN transfer-learning pipeline (schedulers, LR finder, TTA), GPU experiments, error analysis, Gradio demo with Grad-CAM, slides. **Both:** experiment design, analysis, presentation.

References: Rodriguez Bertorello, Sripada, Dendumrongsup, *Amazon Inventory Reconciliation Using AI* (Stanford CS229); Amazon Bin Image Dataset (AWS Open Data); He et al., ResNet (CVPR 2016); Tan & Le, EfficientNet (ICML 2019); Loshchilov & Hutter, SGDR (ICLR 2017); Smith, Cyclical LR (WACV 2017); Selvaraju et al., Grad-CAM (ICCV 2017); Dalal & Triggs, HOG (CVPR 2005).